Start

To start using the strategy camera just drag and drop into the scene Camera Rig prefab inside folder - Universal Strategy Camera -> Prefabs.
Make sure you are using the new Input System and EventSystem is present in the scene. See below.

New Input system

Make sure that you project is using new input system, to add it just open

1. Windows -> Package Manager
2. Select Packages: Unity Registry
3. Find and Import “Input System”

Also don`t forget to change “Active Input Handling”

1. Edit -> Project Settings -> Player
2. Drop down “Other Settings” and find “Active Input Handling”
3. Choose “Input System Package (new)” or “Both”

EventSystem

Make sure that the EventSystem (Gameobject) in your scene is replaced by the new InputSystem Event Module. If you are using the old EventSystem, when you click on EventSystem object you will be prompted to change it by clicking the button in the Inspector

Controls

<u>Mouse/Keyboard:</u>

Left Button - Click
Middle Button - Move
Right Button - Rotate
Scroll - Zoom

WASD, Arrows - Move
E,Q - Rotate
Z,C - Tilt
R,F - Zoom
Shift - SlowMode
` - Reset Camera

<u>Gamepad:</u>

Right Stick - Move
Left Stick - Virtual Mouse Move
Dpad - Rotate
Shoulders - Zoom
Right Trigger, Button South - Click
Left Trigger - SlowMode
Button North - Reset Camera

<u>Touch:</u>

Tap - Click
Pan - Move
Pinch - Zoom
Two Fingers move Horizontally or Vertically - Rotate

Settings

Initial rotation of your camera is determined by rotation of the “Camera Rig” object.
Initial zoom is determined by the Z position of the “Main Camera” object which is a child of “Camera Rig”.

- **Camera Transform -** is a camera that is in use at the moment. It should be child of “Camera Rig” gameobject.
- **Camera Type -** Two options are available: 3D and 2D. Choose based on your game.
- **Virtual Cursor Transform** - is set by default to be “VirtualMouse” Image gameobject located inside “VirtualMouseCanvas”. This visual representation of the mouse will appear when the player is using a gamepad.
- **Follow Transform** - set this field if you want you camera to follow the object

General Sensitivity

- **Slow Speed** - sensitivity of SlowMode activated by Shift on keyboard and Left Trigger on gamepad - slows down zoom, rotation and move.

- **Edge Move Sensitivity** - when the cursor is at the Screen Edge, the camera will move in that direction with this speed.
- **Move Sensitivity -** speed at which camera will move when using the following: WASD, Arrows, LeftStick
- **Button Rotation Sensitivity -** speed of Button rotation: E,Q on Keyboard, DPAD on gamepad
- **Button Zoom Sensitivity** - speed of Button zoom - R,F on Keyboard, Shoulders on gamepad

Gamepad Sensitivity

- **Virtual Mouse Sensitivity** - Sensitivity of virtual mouse controlled by RightStick on Gamepad.

Mouse Sensitivity

- **Mouse Zoom Sensitivity** - Sensitivity of mouse scroll zoom
- **Mouse Rotation Sensitivity -** Sensitivity of mouse rotation triggered by holding the mouse right button and moving the mouse.

Touch Sensitivity

- **Touch Zoom Sensitivity -** Sensitivity of touch zoom. Triggered by pinch.
- **Touch Rotation Sensitivity -** Sensitivity of touch rotation. Triggered by putting two fingers and moving them simultaneously in horizontal or vertical direction.

Zoom Limits

- **Zoom Min** - camera will not be able to zoom in closer than this value
- **Zoom Max -** camera will not be able to zoom out further than this value

Position Limits

- **Boundary Type -** can be a circle with a radius or square with given limits.
- **Square Limits** - if using Boundary Type:Square then set these limit values.

X

W Y
Z

X - Upper Edge, Y - Right edge, Z - Bottom Edge, W - Left Edge

- **Circle Limits** - if using Boundary Type:Circle then set this radius value..

Rotation Limits

- **Rotation Min Max** - will limit horizontal rotation of “Camera Rig” (Y axis rotation). If the value is 0, then no rotation limit will be applied.
- **Tilt Min Max** - - will limit vertical rotation of “Camera Rig” (X axis rotation). If the value is 0, then no tilt limit will be applied.

Technical

- **Lerp Time** - the higher this value the faster camera will apply new values.
- **Floor level** - Up axis position of your ground/terrain object. Usually it is 0.
- **Edge Padding** - Distance from the edge of your screen for moving the camera when the cursor is at the edge.
- **Touch Distance reset** - When doing a click on touch device the finger will move a little, this distance value will ensure that double click and long click will not be reseted.
- **Long Click Duration** - How many seconds a player should press click before it will be considered a long click.
- **Double click Time** - Within how many seconds a player should click twice for it to be considered a double click.

Custom Events and Functions

You can see the usage of custom events and functions in BuildingPlacer script, which allows you to long press on objects that belong to layer “Draggable” and move them around with grid snapping.

Events

onClick(Vector2 Position)

This event is fired when user clicks at a certain position. See BuildingPlacer script for implementation.

onDoubleClick(Vector2 Position)

This event is fired when user double clicks at a certain position. See BuildingPlacer script for implementation.

onLongClick(Vector2 Position)

This event is fired when user long presses at a certain position. See BuildingPlacer script for implementation.

Functions

GetCursorPosition()
Returns current cursor position of mouse when using mouse, virtual mouse when using gamepad. See BuildingPlacer script for implementation.

ClickButtonDown()
Returns true if touch or mouse click was pressed this frame.

ClickButtonPressed()
Returns true if touch or mouse click is pressed right now.

ClickButtonUp()
Returns true if touch or mouse click was released this frame.

HideCursor()
Use this function if you want to hide mouse and virtual mouse cursors. Useful when dragging objects. See BuildingPlacer script for implementation.

ShowCursor()
Use this function if you want to show mouse and virtual mouse cursors. Use when dragging is ended. See BuildingPlacer script for implementation.

TouchMove_OFF()
Use this function when dragging of an object starts. To prevent camera movement on touch devices when dragging an object. Do not forget to turn it back on. See BuildingPlacer script for implementation.

TouchMove_ON()
Use this function when dragging of an object ends. Use after TouchMove_OFF. See BuildingPlacer script for implementation.

PlaneRayCast(Vector3 clickPosition, bool CenterOfScreen = false)
This function performs and returns position of ray on plane.
Optional parameter CenterOfScreen will cast a ray from the center of the screen.

SnapToGrid(Vector2 gridSize, Vector3 position)
This function **returns -** Vector3 snapped position.
Vector2 **gridSize** - size of the grid to which you want to snap the given position.
Vector3 **position** - position which you want to snap to the grid.
See BuildingPlacer script for implementation.